

SOAR Incident Containment Engine

API Testing Report & Documentation

Objective: Test all implemented APIs, verify response outputs, identify observations/issues, and update project documentation with endpoint details, usage examples, and screenshots.

Testing Environment

- Framework: FastAPI
- Testing Tool: Swagger UI
- Base URL: `http://localhost:8000`
- Operating System: Kali Linux

Summary

All implemented API endpoints were tested successfully. Responses were received for all major functions including alert ingestion, dashboard statistics, incident intelligence retrieval, response actions, and reporting endpoints.

GET /

Purpose: Verify API availability.
Result: Passed

POST /alerts

Purpose: Submit security alerts.
Result: Passed

GET /alerts

Purpose: Retrieve stored alerts.
Result: Passed

GET /dashboard/summary

Purpose: Retrieve dashboard statistics.
Result: Passed

GET /dashboard/recent

Purpose: Retrieve recent incidents.
Result: Passed

GET /incident/{id}/intel

Purpose: Retrieve threat intelligence details.
Result: Passed

GET /incidents/high-risk

Purpose: Retrieve high-risk incidents.

Result: Passed

POST /response/isolate-host/{id}

Purpose: Execute host isolation response.

Result: Passed

GET /responses

Purpose: Retrieve response history.

Result: Passed

GET /dashboard/response-metrics

Purpose: Retrieve response metrics.

Result: Passed

GET /dashboard/trends

Purpose: Retrieve dashboard trends.

Result: Passed

Observations Identified During Testing

1. Placeholder/sample values were observed in some recent incident records and should be reviewed.
2. Incident validation for response actions should be verified for non-existent IDs.
3. Duplicate response entries were observed and should be reviewed to determine whether they are expected behavior.
4. Dashboard statistics should be periodically validated against stored alert data.

Screenshots

Screenshot 1

The screenshot displays a web browser window with the Swagger UI for a FastAPI application. The browser's address bar shows the URL `localhost:8000/docs/#/`. The Swagger UI interface is divided into several sections:

- Server response:** This section shows the details of a successful response (200). The response body is a JSON object: `{ "message": "SOAR Incident Containment Engine API Running" }`. The response headers are: `content-length: 58`, `content-type: application/json`, `date: Fri, 12 Jun 2020 14:59:35 GMT`, and `server: uvicorn`. A "Download" button is visible next to the response body.
- Responses:** This section lists the available API endpoints. The first entry is a 200 response for the `/alerts` endpoint, described as "Successful Response". The media type is set to `application/json`. The example value is `"string"`.
- GET /alerts:** This section shows the details for the `GET /alerts` endpoint, which is described as "Get Alerts".
- POST /alerts:** This section shows the details for the `POST /alerts` endpoint, which is described as "Receive Alert".

The interface includes a "Try it out" button at the bottom right of the `POST /alerts` section.

Screenshot 2

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/alerts' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "src_ip": "string",
    "severity": "string",
    "event_type": "string",
    "timestamp": "string"
  }'
```

Request URL

http://localhost:8000/alerts

Server response

Code	Details
200	Response body <pre>{ "status": "success", "message": "Alert stored successfully", "risk_score": 10, "threat": false }</pre> Response headers <pre>content-length: 89 content-type: application/json date: Fri, 12 Jun 2026 15:04:44 GMT server: uvicorn</pre>

Responses

Screenshot 3

mass assi... Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

Curl

```
curl -X 'GET' \
  'http://localhost:8000/' \
  -H 'accept: application/json'
```

Request URL

http://localhost:8000/

Server response

Code	Details
200	Response body <pre>{ "message": "SOAR Incident Containment Engine API Running" }</pre> Response headers <pre>content-length: 58 content-type: application/json date: Fri, 12 Jun 2026 14:59:35 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

Screenshot 4

Controls Accept header.

Example Value | Schema

```
"string"
```

422 Validation Error

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Screenshot 5

Curl

```
curl -X 'GET' \
  'http://localhost:8000/dashboard/summary' \
  -H 'accept: application/json'
```

Request URL

http://localhost:8000/dashboard/summary

Server response

Code	Details
200	Response body <pre>{ "total_alerts": 3, "high_alerts": 1, "medium_alerts": 0, "low_alerts": 0 }</pre> Response headers <pre>content-length: 67 content-type: application/json date: Fri, 12 Jun 2026 18:58:29 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

Screenshot 6

Request URL

http://localhost:8000/dashboard/recent

Server response

Code	Details
200	Response body <pre>{ "id": 3, "src_ip": "string", "severity": "string", "event_type": "string", "timestamp": "string", "status": "OPEN" }, { "id": 2, "src_ip": "string", "severity": "string", "event_type": "string", "timestamp": "string", "status": "OPEN" }, { "id": 1, "src_ip": "192.168.1.100", "severity": "high", "event_type": "malware", "timestamp": "2026-06-09 21:38:00", "status": "OPEN" } }</pre> Download Response headers <pre>content-length: 338 content-type: application/json date: Fri, 12 Jun 2026 18:59:10 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Screenshot 7

Curl

```
curl -X 'GET' \
  'http://localhost:8000/incident/1/intel' \
  -H 'accept: application/json'
```

Request URL

http://localhost:8000/incident/1/intel

Server response

Code	Details
200	Response body <pre>{ "incident_id": 1, "src_ip": "192.168.1.100", "risk_score": 10, "threat": false }</pre> Download Response headers <pre>content-length: 73 content-type: application/json date: Fri, 12 Jun 2026 19:04:10 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Screenshot 8

Responses

Code

Description

200

Successful Response

No links

Media type

application/json

Controls Accept header.

Example Value

Schema

"string"

422

Validation Error

No links

Media type

application/json

Controls Accept header.

Example Value

Schema

{
 "detail": [
 {
 "loc": [
 "string",
 0
],
 "msg": "string",
 "type": "string",
 "input": "string",
 "ctx": {}
 }
]
}

Completed — 90 bytes

Show all downloads

Screenshot 9

curl -X 'GET' \
 'http://localhost:8080/incidents/high-risk' \
 -H 'accept: application/json'

Request URL

http://localhost:8080/incidents/high-risk

Server response

Code

Details

200

Response body

[
 {
 "id": 1,
 "src_ip": "192.168.1.100",
 "severity": "high",
 "status": "OPEN"
 }
]

Response headers

content-length: 69
content-type: application/json
date: Fri, 12 Jun 2020 19:05:16 GMT
server: uvicorn

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header.

Example Value

Schema

"string"

Download

Screenshot 10

```
curl -X 'POST' \
  'http://localhost:8080/response/isolate-host/101' \
  -H 'accept: application/json' \
  -d ''
```

Request URL

http://localhost:8080/response/isolate-host/101

Server response

Code	Details
200	Response body <pre>{ "message": "Host isolated successfully" }</pre> Download Response headers <pre>content-length: 40 content-type: application/json date: Fri, 12 Jun 2026 19:07:15 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema <pre>"string"</pre>	No links

Screenshot 11

422

Validation Error

No links

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


Screenshot 12

Responses

Curl

curl -X 'GET' \ 'http://localhost:8000/responses' \ -H 'accept: application/json'

Request URL

http://localhost:8000/responses

Server response

Code

Details

200

Response body

```
{
  {
    "id": 1,
    "incident_id": 1,
    "action_type": "BLOCK_IP",
    "status": "SUCCESS",
    "timestamp": "2026-06-11 07:50:50.674872"
  },
  {
    "id": 2,
    "incident_id": 1,
    "action_type": "ISOLATE_HOST",
    "status": "SUCCESS",
    "timestamp": "2026-06-11 07:59:24.174507"
  },
  {
    "id": 3,
    "incident_id": 100,
    "action_type": "BLOCK_IP",
    "status": "SUCCESS",
    "timestamp": "2026-06-12 15:06:09.697440"
  },
  {
    "id": 4,
    "incident_id": 100,
    "action_type": "BLOCK_IP",

```

Download

Response headers

Screenshot 13

Curl

curl -X 'GET' \ 'http://localhost:8000/dashboard/response-metrics' \ -H 'accept: application/json'

Request URL

http://localhost:8000/dashboard/response-metrics

Server response

Code

Details

200

Response body

```
{
  "total_actions": 6,
  "blocked_ips": 3,
  "isolated_hosts": 3
}
```

Download

Response headers

```
content-length: 54
content-type: application/json
date: Fri, 12 Jun 2026 19:10:04 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value

Schema

```
"string"
```

Screenshot 14

Curl

```
curl -X 'GET' \
  'http://localhost:8080/dashboard/trends' \
  -H 'accept: application/json'
```

Request URL

http://localhost:8080/dashboard/trends

Server response

Code	Details
200	Response body <pre>{ "high": 1, "medium": 0, "low": 0 }</pre> Download Response headers <pre>content-length: 29 content-type: application/json date: Fri, 12 Jun 2026 19:10:38 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string"	No links

Screenshot 15

Schemas

Alert ^ Collapse all object

src_ip* string

severity* string

event_type* string

timestamp* string

HTTPValidationError ^ Collapse all object

detail > Expand all array<object>

ValidationError ^ Collapse all object

loc* > Expand all array<(string | integer)>

msg* string

type* string

input any

ctx object

Conclusion

The API testing activity was completed successfully. All available endpoints responded correctly during testing. Observations identified during testing have been documented for future review and improvement. The project is operational and the implemented APIs are functioning as expected.